

Day 2: SQL Notes from Basic to Advanced

TODAYS TOPICS: SQL Database Basics

- What is a Database?
- SQL Data Types
- Creating and Modifying Tables
- Inserting Data into Tables

2. SQL Database Basics

2.1 What is a Database?

- A database is an organized collection of data stored electronically.
- It helps in managing and retrieving data efficiently.
- Databases are used in various applications like websites, banking systems, and more.
- Databases help in efficiently managing large amounts of data and can be queried using SQL.

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2.2 SQL Data Types

SQL data types define the nature of the data that can be stored in each column of a table. Here are some common SQL data types:

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1. Numeric Data Types

- **INT**: Stores whole numbers (e.g., 1, 100). Suitable for integers.
- **FLOAT**: Stores floating-point numbers (e.g., 3.14, 2.71). Suitable for numbers with decimals.
- **DECIMAL**: Stores exact numeric values with a fixed number of decimal places (e.g., 123.45). Useful for financial calculations.

2. Character and String Data Types

- **CHAR(n)**: Stores fixed-length strings of text. The length is specified by n (e.g., CHAR(10) stores strings with exactly 10 characters, padding with spaces if needed).
- **VARCHAR(n)**: Stores variable-length strings of text. The maximum length is specified by n (e.g., VARCHAR(255) can store strings up to 255 characters long).

3. Date and Time Data Types

- **DATE:** Stores dates in the format YYYY-MM-DD (e.g., 2024-08-05).
- **TIME:** Stores time of day (e.g., 14:30:00 for 2:30 PM).
- **DATETIME:** Stores both date and time (e.g., 2024-08-05 14:30:00).

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4. Other Data Types

- **BOOLEAN:** Stores TRUE or FALSE values. Useful for flags or binary choices.
- **BLOB:** Stores large binary objects, such as images or files. Suitable for storing binary data.

2.3 Creating and Modifying Tables

Creating Tables

To create a new table, use the CREATE TABLE statement. You specify the table name and define its columns along with their data types.

Note: For information on how to create tables, refer to Day 1.

Syntax:

```
CREATE TABLE table_name (  
    column1 data_type constraint,  
    column2 data_type constraint,  
    ...  
);
```

Example:

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    FirstName VARCHAR(50),  
    LastName VARCHAR(50),  
    HireDate DATE  
);
```

In this example:

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- EmployeeID is an integer and serves as the primary key.
- FirstName and LastName are variable-length text fields.
- HireDate stores the date the employee was hired.

2.3.2) Modifying Tables

To modify an existing table, use the ALTER TABLE statement. This can involve adding, deleting, or modifying columns and constraints.

1. Adding Columns

Syntax to add a new column to an existing table:

```
ALTER TABLE table_name  
ADD column_name data_type;
```

Example:

```
ALTER TABLE Employees  
ADD Email VARCHAR(100);
```

This adds an Email column to the Employees table.

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2. Modifying Columns

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Syntax to change the data type or constraints of an existing column:

```
ALTER TABLE table_name  
MODIFY COLUMN column_name new_data_type;
```

Example:

```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

This changes the LastName column to allow up to 100 characters.

3. Dropping Columns

To remove a column from a table.

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Syntax:

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```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

Example:

```
ALTER TABLE Employees  
DROP COLUMN Email;
```

This removes the Email column from the Employees table.

2.4 Inseting Data into Tables

To insert data into a table in SQL, you use the INSERT INTO statement. Here's a simple guide on how to do it:

Inserting Data into a Table

Syntax:

```
INSERT INTO table_name (column1, column2, ... )  
VALUES (value1, value2, ... );
```

Example: Assuming you have a table named Employees with columns EmployeeID, FirstName, LastName, and HireDate:

```
INSERT INTO Employees (EmployeeID, FirstName, LastName, HireDate)  
VALUES (1, 'John', 'Doe', '2024-08-05');
```

In this example:

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- EmployeeID is set to 1.
- FirstName is set to 'John'.
- LastName is set to 'Doe'.
- HireDate is set to '2024-08-05'.

Inserting Multiple Rows

You can also insert multiple rows in a single INSERT statement by separating each set of values with a comma.

Syntax:

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...),
       (value1, value2, ...),
       ... ;
```

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Example:

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```
INSERT INTO Employees (EmployeeID, FirstName, LastName, HireDate)
VALUES (2, 'Jane', 'Smith', '2024-08-06'),
       (3, 'Alice', 'Johnson', '2024-08-07');
```

Inserting Data with Default Values

If some columns have default values or allow NULL, you can omit those columns from the INSERT statement.

Syntax:



```
INSERT INTO table_name (column1, column2, ...)  
SELECT column1, column2, ...  
FROM another_table  
WHERE condition;
```

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Example:

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```
INSERT INTO Employees (EmployeeID, FirstName, LastName, HireDate)  
SELECT EmployeeID, FirstName, LastName, HireDate  
FROM NewEmployees  
WHERE HireDate > '2024-01-01';
```

This command inserts rows into the Employees table that match the criteria from the NewEmployees table.

Using these INSERT INTO commands, you can add new data to your tables efficiently.

Day 3:Sql Constraints

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